

FLS Realized-Geography Entropy Balancing

The procedure estimates a public-data analog of each FLS region-year’s geographic composition. It does not recover NASS’s confidential sample or survey weights.

The baseline prior q_i is each OEWS area’s share of the region’s Census hired-worker frame analog. County mass is allocated to OEWS areas using within-county township shares. Areas remain in the frame regardless of whether an OEWS wage is observed.

The calibration targets are published FLS worker counts crossed by reference quarter and duration: 150 days or more versus fewer than 150 days. Counts are normalized into a joint composition τ . Public area analogs combine strict QCEW agricultural employment, QWI fills and persistence, and Census duration shares. The resulting area compositions are converted to prior-standardized orthonormal Helmert contrasts, avoiding an arbitrary omitted cell.

For a prior p , the estimator solves

$$\min_{\mathbf{w}} D_{KL}(\mathbf{w} \parallel \mathbf{p}) + \frac{\rho}{2} \|\tilde{\mathbf{Z}}' \mathbf{w} - \tilde{\tau}\|_2^2$$

subject to $w_i \geq 0$ and $\sum_i w_i = 1$. The KL term keeps the recovered distribution near the frame prior; the second term rewards agreement with the FLS quarter-duration moments. Larger ρ pursues those moments more strongly. The primary value is $\rho = 0.10$.

The deterministic center uses $p = q$. To represent plausible realized sampling variation, the procedure also draws

$$\mathbf{q}^{(b)} \sim \text{Dirichlet}(\kappa \mathbf{q}), \quad \kappa = m \left(\frac{1}{\sum_i q_i^2} \right),$$

and solves the same entropy problem for every draw. The primary specification uses $m = 10,999$ draws, and seed 20260726; fixed m and ρ alternatives provide sensitivity checks.

Wages, AEW values, first-stage estimates, and outcomes never enter fitting. OEWS and FLS wages are introduced only afterward for held-out descriptive validation. Reported draw percentiles are simulation envelopes for the predeclared ensemble.